

NETWORK OPTIMIZATION GUIDE

End-to-End Latency Optimization for Gaming & Streaming

Platform: JIO Single NAT Home Network | Hardware: Jio ONT + Tenda F3
Games Tested: CS2 | Valorant | OS: Windows 11 + MacOS Sequoia

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1. Introduction & Purpose

Most gamers optimize their PC hardware — GPU, RAM, CPU — and completely ignore the one thing that directly affects in-game responsiveness: the network stack between their machine and the game server. This guide documents a real, hands-on network optimization performed on a JioFibre home network to squeeze out every last millisecond of latency for competitive gaming.

The approach is structured and methodical — baseline first, then systematic changes, then verification. Every tweak is logged with timestamps, test results, and delta comparisons. No guesswork, no placebo tweaks. Just data.

Primary Objectives:

- Eliminate bufferbloat and reduce ping spikes under load
- Configure OS-level QoS to prioritize CS2 and Valorant UDP traffic
- Lock stable IP addressing to prevent mid-game connection drops
- Verify MTU, DNS resolution speed, and TCP stack settings
- Document all changes with before/after measurements for full transparency

2. Environment & Hardware Setup

Hardware

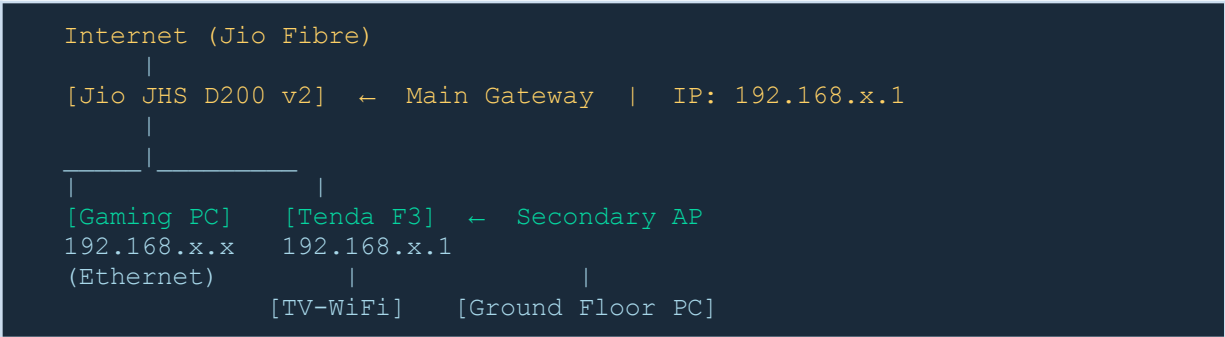
Device	Model	Role	Notes
User's Gaming PC	Custom Build	Primary Client	Ethernet, Windows 11 Home
ISP Router / ONT	Jio JHS D200 v2	Main Gateway	ISP-locked, view-only admin
Secondary Router	Tenda F3 N300	AP for other devices	No OpenWrt / DD-WRT support
Apple MacBook Air	MacBook Air M4	iperf3 Test Client	Wi-Fi only, used for LAN testing

Software & Tools Used

Tool	Purpose	Platform
iperf3	LAN throughput testing (server ↔ client)	Windows + Mac
ping (CMD)	Latency baseline and packet loss	Windows
nslookup	DNS verification	Windows
Waveform Bufferbloat Test	Bufferbloat grade + loaded latency	Browser
fast.com	WAN speed test (download/upload/latency)	Browser
TCP Optimizer (SG)	Windows TCP stack fine-tuning	Windows
Windows Registry (regedit)	DSCP QoS policy for CS2/Valorant	Windows

Network Topology

The physical layout of our network is a single NAT setup — no double NAT — which is ideal for gaming. The Gaming PC connects directly via Ethernet to the Jio ONT. The Tenda F3 is wired off the ONT as an access point for Wi-Fi devices. It's always important to understand your network topology which will lead to better optimization, fix to a certain device and better results.



NAT Type: Single NAT ☒ | No Double NAT: Checked ☒ — Clean topology, no extra hop overhead.

3. Phase 1 — Baseline Testing

All measurements were taken before any changes were made. This is the ground truth used to calculate every delta in the results section.

3.1 Ping Baseline — DNS Comparison

Before making any changes, 100-packet ICMP ping tests were run to both Google DNS and Cloudflare DNS. The reason we switched to Cloudflare was simple: the test results proved it — Cloudflare delivered a 4ms average vs Google's 43ms average from our location (West Bengal, India). That is a 10× difference in DNS resolution speed. Data made the decision, not assumption.

Target	Min Ping	Avg Ping	Max Ping	Packet Loss	Verdict
Google DNS (8.8.8.8)	43ms	45ms	228ms 🚩	0%	Spike — too slow
Cloudflare (1.1.1.1)	4ms ✅	5ms ✅	11ms	0%	Winner — 10× faster ✅

⚠️ **Red Flag:** Google DNS max spike of 228ms indicates possible bufferbloat under load. Cloudflare's 4ms average vs Google's 43ms average made the DNS switch a clear data-driven decision — not a preference.

3.2 LAN Throughput — iperf3

A 30-second iperf3 test was run between the Gaming PC (Ethernet, server) and MacBook Air M4 (Wi-Fi, client) to measure LAN performance and isolate interference sources.

Metric	Value	Status
Server (Gaming PC)	Ethernet connection	Stable ✅
Client (MacBook Air M4)	Wi-Fi connection	Variable
Test Duration	30 seconds	
Peak Bandwidth	273 Mbps	Healthy
Average Bandwidth	220 Mbps	Good ✅
Retransmits	0	Zero packet loss ✅
Speed Dips (sec 3-9, 24-25)	182 Mbps dip	Wi-Fi interference ⚠️

Conclusion: The Ethernet side is rock solid. The dips are caused by Wi-Fi airtime sharing on the Mac client — not the LAN infrastructure. LAN itself is clean. A more stable Wi-Fi device could have helped here for Mac device.

3.3 WAN Speed Test – fast.com

Metric	Value	Assessment
Download Speed	150 Mbps	Good
Upload Speed	170 Mbps	Good
Idle Latency	3ms	Excellent ✓
Loaded Latency	64ms 🚨	Bufferbloat confirmed!
Bufferbloat Delta	+61ms under load	21× spike — SQM needed

3.4 Bufferbloat Test – Waveform

Metric	Baseline Value	Status
Grade	B	Acceptable but not great
Idle Latency	3.95ms	✓ Clean
Download Latency Spike	+41ms	🚨 Bufferbloat
Upload Latency Spike	0ms	✓ Clean
Peak Download Ping	82.4ms	🚨 Too high for competitive
Low Latency Gaming	FAIL ✖	4ms over threshold

🚨 **Root Cause:** Download-side bufferbloat only. Upload is clean — meaning the buffer is inside the ISP modem, not the router. SQM on the download lane is the fix.

4. Optimization Phases — What Was Done & Why

1 Firmware Flash Assessment — Ruled Out

Before touching any settings, firmware compatibility was checked for both routers against DD-WRT and OpenWrt databases. The goal was to unlock SQM (Smart Queue Management) — the most effective bufferbloat fix.

- Tenda F3: Only 4MB flash storage. Minimum for DD-WRT/OpenWrt is 8MB. No community builds exist. Ruled out.
- Jio JHS D200 v2: ISP-locked ONT. Cannot be reflashed without ISP reprovisioning. Ruled out.

Result: Both firmware paths abandoned. Proceeding with software-side optimizations only.

2 DNS Optimization — Switched to Cloudflare 1.1.1.1

Jio's default ISP DNS (reliance.reliance) was tested and proved slow. The baseline ping results were the deciding factor: Cloudflare returned an average of 4ms vs Jio DNS at 43ms — a 10× difference from our West Bengal location. Every game session starts with a DNS lookup. Switching to the faster resolver means less time waiting before your first packet even leaves the machine.

Changes Applied

- Configured Cloudflare DNS on Windows Ethernet adapter (IPv4 + IPv6)
- Primary DNS: 1.1.1.1 — Secondary DNS: 1.0.0.1
- IPv6 also configured to prevent fallback to Jio ISP DNS on the IPv6 path
- Verified via nslookup — Cloudflare responding correctly at one.one.one.one

 **Why adapter-level?:** DNS set on the Jio router admin panel did not apply correctly — Jio's dashboard doesn't properly push DNS for the IPv6 path. Setting it directly on the Windows adapter is more reliable and survives router reboots.

3 Static IP Assignment — Locked Gaming PC

DHCP-assigned IPs can change between sessions, which breaks QoS rules, port forwarding targets, and iperf3 server addresses. Locking to a static IP eliminates this variable permanently.

DHCP reservation on the Jio admin panel was view-only (ISP restriction). Static IP was therefore set directly on the Windows Ethernet adapter — more reliable since it survives router reboots without needing the router to honor any reservation.

Configuration Applied

Setting	Value
IP Address	192.168.x.x (your specific LAN address — masked)
Subnet Mask	255.255.255.0
Default Gateway	192.168.x.1 (your router)
Primary DNS	1.1.1.1 (Cloudflare)
Secondary DNS	1.0.0.1 (Cloudflare fallback)
Method	Windows 11 Adapter — manual static assignment

4 MTU Verification — Confirmed Optimal

MTU (Maximum Transmission Unit) is the largest packet your connection can send in one go without splitting it. If MTU is wrong, every oversized packet gets fragmented and re-assembled, adding latency on every single frame. For Ethernet + most ISPs, 1500 bytes is correct.

Test Method — Ping with Don't Fragment Flag

```
ping -f -l 1472 8.8.8.8 ← PASS (packet went through intact)
ping -f -l 1473 8.8.8.8 ← FRAGMENTED (exceeds limit — MTU ceiling found)
```

Result: MTU = 1500 confirmed. 1472 payload + 28 bytes IP/ICMP headers = 1500 exactly. No changes needed.

5 QoS — DSCP 46 Priority for CS2 & Valorant

QoS (Quality of Service) tells your OS and router which traffic matters most. DSCP value 46 (Expedited Forwarding) is the highest real-time priority class in the DiffServ standard (RFC 2474). Setting this for game processes means when your internet is busy, game UDP packets jump the queue.

Windows 11 Home does not have Group Policy Editor (gpedit.msc). QoS was applied via direct Windows Registry entries — the documented fallback that works identically.

Registry Commands — CS2 (same block code for Valorant, swap cs2.exe)

```
reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows\QoS\CS2" /v "Version" /t REG_SZ /d "1.0" /f
reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows\QoS\CS2" /v "Application Name" /t REG_SZ /d "cs2.exe" /f
reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows\QoS\CS2" /v "Protocol" /t REG_SZ /d "UDP" /f
reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows\QoS\CS2" /v "DSCP Value" /t REG_SZ /d "46" /f
reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows\QoS\CS2" /v "Throttle Rate" /t REG_SZ /d "-1" /f
```

For Valorant replace CS2 / cs2.exe with Valorant / VALORANT-Win64-Shipping.exe. Protocol is UDP for both games.

Verification After Restart

```
reg query "HKLM\SOFTWARE\Policies\Microsoft\Windows\QoS\CS2"
```

Should return all 5 values. If it does, the policy survived the reboot and is active.

6 TCP Stack Tuning – TCP Optimizer

Windows 11's default TCP stack is built for general use, not low-latency gaming. TCP Optimizer lets you fine-tune how packets are queued, how congestion is handled, and how much buffer Windows pre-allocates.

Setting	Before	After	Why
Speed Slider	10 Mbps	100+ Mbps	Tells Windows 11 the actual connection capacity
TCP Auto-Tuning	Normal	Restricted	Prevents over-buffering on download
Congestion Control	CUBIC	CTCP	CTCP is better for high-bandwidth-delay connections
ECN Capability	Disabled	Enabled	Signals congestion without dropping packets
TCP Chimney	Disabled	Enabled*	*Tested — caused upload spikes, reverted

⚠ **TCP Chimney Note:** Enabling TCP Chimney caused upload latency to spike above 220ms on the Realtek Gaming 2.5GbE NIC. It was reverted. NIC behavior varies — always benchmark before keeping it on.

7 Port Forwarding – Blocked by ISP

Port forwarding was attempted for CS2 (UDP 27015–27030) and Valorant (UDP 7000–8000) on the Jio admin panel. The Services dropdown only shows predefined options (HTTP, SIP, FTP) — no custom port entry field. Blocked by ISP-side restrictions.

Impact: Without explicit port forwarding, NAT type stays at default. Since this is already Single NAT (no double NAT), the practical gaming impact is low — but OPEN NAT would require either a third-party router or ISP cooperation.

5. Results — Before vs After

All post-optimization measurements were taken under identical conditions to the baseline. Deltas are objective numbers, not estimates.

5.1 Ping Results

Metric	Before	After	Delta
Google Max Spike	228ms	36ms	-192.0ms
Google Avg	45ms	32ms	-13.0ms
Google Min	43ms	32ms	-11.0ms
Cloudflare Max	11ms	11ms	0.0ms
Cloudflare Avg	5ms	5ms	0.0ms

🔥 **Biggest Win:** Google DNS max spike eliminated: 228ms → 36ms. That is 192ms of jitter gone — nearly a fifth of a second of random latency removed.

5.2 Bufferbloat Results

Metric	Before	After	Delta
Idle Latency	3.95ms	2.78ms	-1.2ms
Download Spike	41ms	34ms	-7.0ms
Peak Ping	82.4ms	57.49ms	-24.9ms
Upload Spike	0ms	1ms	+1.0ms

Metric	Baseline	Post-Optimization	Delta
Grade	B	B	Same — ISP hardware ceiling
Idle Latency	3.95ms	2.78ms	-1.17ms (-30%) ✓
Download Spike	41ms 🚨	34ms	-7ms (-17%)
Peak Download Ping	82.4ms	57.49ms	-24.9ms (-30%) ✓
Low Latency Gaming	FAIL ✖	FAIL ✖	4ms over threshold — ISP buffer limit

🧠 **Why Still Failing:** Gaming PASS threshold is <30ms download spike. We are at 34ms — just 4ms away. The remaining buffer is inside Jio's ONT hardware, inaccessible from Windows or admin panel. SQM on a compatible router is the fix.

5.3 Summary Scorecard

Optimization	Status	Impact
Cloudflare DNS	✓ Active	10× faster DNS resolution vs ISP default
Static IP	✓ Locked	Stable connection — no mid-game IP changes
DSCP 46 QoS (CS2 + Valorant)	✓ Active	OS-level UDP traffic priority
MTU 1500	✓ Confirmed	Zero fragmentation overhead
TCP Optimizer	✓ Applied	Improved congestion control + buffer sizing
Bufferbloat / SQM	⚠ Partial	17% improvement — blocked by ISP hardware
Firmware Flash	✗ Blocked	Hardware incompatible — insufficient flash memory
Port Forwarding	✗ Blocked	ISP admin panel restriction — no custom ports

6. Post-Optimization Verification Checklist

After a system restart, run these three checks to confirm all settings survived the reboot.

Check 1 — QoS Registry Entries

Open CMD as Administrator and run:

```
reg query "HKLM\SOFTWARE\Policies\Microsoft\Windows\QoS\CS2"
```

Expected: All 5 values visible (Version, Application Name, Protocol, DSCP Value, Throttle Rate).

Check 2 — DNS Still Active

```
nslookup google.com
```

Expected: Response from one.one.one.one (Cloudflare). If you see a Jio/reliance address, re-apply DNS on the adapter.

Check 3 — Static IP Locked

```
ipconfig
```

Expected: Gaming PC shows the same static IP from Phase 3. If it shows a different DHCP address, re-apply the static IP on the Windows adapter.

7. Challenges, Blockers & Limitations

Not everything could be fixed. Some limitations came from ISP-locked hardware, some from incompatible firmware, some from Windows edition restrictions. Below is a clear breakdown — what was blocked, what we would have done, and what the result would have been.

7.1 What We Could Not Do — Visual Breakdown

Firmware Flash (Tenda F3 + Jio ONT)

What was blocked	What we would have done	Result if it worked
Tenda F3 has only 4MB flash — minimum for DD-WRT/OpenWrt is 8MB. No builds exist. Jio ONT is ISP-locked and cannot be reflashed without ISP reprovisioning.	We would flash OpenWrt on a compatible router and enable SQM (Smart Queue Management) — a traffic shaping algorithm that limits the buffer depth to prevent bloat in real time.	Bufferbloat grade would jump from B → A. Low Latency Gaming test would flip from FAIL → PASS. Download spike would drop from 34ms to under 10ms.

Port Forwarding via Jio Admin Panel

What was blocked	What we would have done	Result if it worked
Jio's admin portal only allows predefined services (HTTP, SIP, FTP) in the port forwarding dropdown. Custom port ranges for CS2 (UDP 27015–27030) and Valorant (UDP 7000–8000) cannot be entered.	We would add custom port forward rules pointing CS2 and Valorant UDP ranges directly to the Gaming PC's static IP, creating a permanent open path through the NAT.	NAT type would upgrade from Moderate/Strict to OPEN NAT — better matchmaking, lower connection latency to game servers, and fewer mid-game disconnects in competitive lobbies.

Router-Level DNS (Jio Dashboard)

What was blocked	What we would have done	Result if it worked
Changing DNS on the Jio admin panel didn't apply properly — the router continued using Jio's ISP DNS on the IPv6 path, bypassing the IPv4 config.	With a proper third-party router, we would set Cloudflare DNS (1.1.1.1) globally at the router level so every device on the network automatically uses the faster resolver without any per-device setup.	All connected devices — gaming PC, phones, TV — would benefit from faster DNS resolution. No manual configuration needed on each device individually.

Group Policy QoS (Windows 11 Home)

What was blocked	What we would have done	Result if it worked
Windows 11 Home does not include Group Policy Editor (gpedit.msc), which is the standard GUI for DSCP QoS policy. Registry method was used as workaround.	On Windows 11 Pro/Enterprise, we would configure QoS directly through gpedit.msc with a clean UI — easier to manage, verify, and update without touching the Registry.	Functionally identical result — DSCP 46 is applied either way. The difference is management simplicity and audit visibility. Registry method works, just less user-friendly.

7.2 ISP Hardware Restrictions

- Jio JHS D200 v2 admin panel is view-only for DHCP reservations — static IP had to be set at the Windows 11 adapter level instead
- Port forwarding blocked — Services dropdown has no custom port entry, only predefined services (HTTP, SIP, FTP)
- DNS change on Jio dashboard did not apply — router used IPv6 path, bypassing IPv4 config

7.3 Firmware Incompatibility

- Tenda F3 has only 4MB flash storage. DD-WRT and OpenWrt both require a minimum of 8MB. No community builds exist for this model.
- Jio ONT cannot be reflashed — ISP-managed hardware with proprietary firmware locked to the MAC address provisioned at installation

8. Upgrade Path — Breaking the ISP Ceiling

The remaining 34ms download bufferbloat cannot be fixed in software. The buffer lives inside Jio's ONT. The only real fix is SQM (Smart Queue Management) on an OpenWrt-compatible third-party router placed between the Jio ONT and your gaming PC.

Here are 5 routers available in India today that would unlock full control — SQM, custom DNS, real port forwarding, VLAN, and complete QoS management.*

8.1 Recommended Routers — Available in India

All routers below support OpenWrt or equivalent custom firmware, enabling SQM for full bufferbloat control, proper port forwarding, and custom DNS at router level.

Budget	Router	Wi-Fi	SQM / OpenWrt	Where to Buy
Under Rs. 2,500	TP-Link TL-WR940N N450	Wi-Fi 4 (N)	DD-WRT only — basic	Amazon.in / Flipkart
Around Rs. 3,500	TP-Link Archer C6 AC1200	Wi-Fi 5 (AC)	OpenWrt — good SQM	Amazon.in / Flipkart
Around Rs. 4,500	TP-Link Archer AX55 AX1800	Wi-Fi 6 (AX)	OpenWrt — full SQM	Amazon.in / Flipkart / Croma
Around Rs. 6,000	Asus RT-AX58U AX3000	Wi-Fi 6 (AX)	OpenWrt + Merlin — best QoS	Amazon.in / Asus India
Rs. 6,000+	GL.iNet Beryl AX MT3000	Wi-Fi 6 (AX)	Ships with OpenWrt — zero setup	Amazon.in



How to set it up: Place any of these routers between the Jio ONT's LAN port and your PC. Enable SQM on the WAN interface with your actual speeds. Bufferbloat grade jumps from B to A, Low Latency Gaming flips to PASS.

9. Key Technical Learnings

- Bufferbloat is ISP-side when download spikes but upload is clean — the buffer is in the modem, not the router or OS.
- DSCP 46 (Expedited Forwarding) is the highest QoS class in DiffServ. For real-time UDP gaming traffic, this is the correct value.
- MTU discovery via DF-set ping needs no extra tools — works on any Windows machine, takes 30 seconds.
- DNS resolution speed varies massively by location. Cloudflare's 4ms vs Jio DNS 43ms from West Bengal — the test data made the decision, not preference.
- TCP Chimney Offload causes latency anomalies on certain NICs. Always benchmark after enabling — never assume.
- Windows 11 Home blocks Group Policy — the Registry is a fully supported and reliable fallback for QoS DSCP configuration.
- Static IP at adapter level is more reliable than DHCP reservation for gaming — no router cooperation required.

10. References

Resource	Purpose	URL / Source
Waveform Bufferbloat Tool	Bufferbloat grade + loaded latency	waveform.com/tools/bufferbloat
SpeedGuide TCP Optimizer	Windows TCP stack tuning	speedguide.net/downloads.php
DD-WRT Router Database	Firmware compatibility check	dd-wrt.com/site/support/router-database
OpenWrt Table of Hardware	Router compatibility research	openwrt.org/toh/start
Cloudflare 1.1.1.1 DNS	Public DNS resolver	1.1.1.1
GL.iNet Official Store	GL-MT3000 Beryl AX India	amazon.in (GL.iNet store)
RFC 2474	DSCP / DiffServ definition	IETF RFC 2474
RFC 4774	ECN (Explicit Congestion Notification)	IETF RFC 4774
iperf3 Documentation	Network throughput testing	software.es.net/iperf
Microsoft Docs — QoS Policy	Windows QoS registry implementation	docs.microsoft.com

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